

Worked examples for Lecture 20: Self-gravitation and rotation

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These examples accompany Lecture 20. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory. Throughout we take $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, and for the Earth a mass $M = 5.97 \times 10^{24} \text{ kg}$, a mean radius $b = 6371 \text{ km}$ and a rotation rate $\Omega = 7.29 \times 10^{-5} \text{ s}^{-1}$.

Example 1: The homogeneous sphere. A sphere of radius b has uniform density ρ and total mass $M = \frac{4}{3}\pi\rho b^3$. Solve Poisson's equation to find the gravitational potential ϕ and acceleration $\mathbf{g}$ inside and outside the sphere. Compute the gravitational binding energy $\mathcal{V}_g$ in two ways, using both expressions obtained in the lecture, and show that $\mathcal{V}_g = -\frac{3}{5}GM^2/b$. Evaluate the result for the Earth and compare it with the energy released by the largest earthquakes.

Solution. We take the reference body to be the sphere itself, so that the motion is the identity, the spatial and referential densities coincide, and $\zeta = \phi$, $\gamma = \mathbf{g}$. By symmetry ϕ depends only on $r = \|\mathbf{x}\|$, and Poisson's equation $\nabla^2\phi = 4\pi G\rho$ reads

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = \begin{cases} 4\pi G\rho, & r < b, \\ 0, & r > b. \end{cases} \quad (1)$$

Integrating twice, the solution that is regular at the origin and tends to zero at infinity is

$$\phi(r) = \begin{cases} \frac{2}{3}\pi G\rho r^2 + A, & r < b, \\ -\frac{C}{r}, & r > b, \end{cases} \quad (2)$$

with A and C constants. Since there are no surface masses, the lecture's continuity conditions require both ϕ and $g_r = -\frac{d\phi}{dr}$ to be continuous at $r = b$. Continuity of the radial derivative gives $\frac{4}{3}\pi G\rho b = C/b^2$, hence $C = \frac{4}{3}\pi G\rho b^3 = GM$, while continuity of ϕ then gives $A = -GM/b - \frac{2}{3}\pi G\rho b^2 = -GM/b - GM/(2b)$, where we have used $\rho = 3M/(4\pi b^3)$. Collecting terms,

$$\phi(r) = \begin{cases} -\frac{GM}{2b^3} (3b^2 - r^2), & r \leq b, \\ -\frac{GM}{r}, & r \geq b, \end{cases} \quad g_r(r) = \begin{cases} -\frac{GM}{b^3} r = -\frac{4}{3}\pi G\rho r, & r \leq b, \\ -\frac{GM}{r^2}, & r \geq b. \end{cases} \quad (3)$$

The exterior field is that of a point mass M at the origin, while inside the sphere the acceleration grows linearly with radius, as only the mass interior to r contributes. The potential at the centre is $-\frac{3}{2}GM/b$, one and a half times its surface value.

For the binding energy we first use the volume form $\mathcal{V}_g = \frac{1}{2} \int_{M_t} \varrho \phi \, d^3\mathbf{y}$ from the lecture, where the integral extends only over the body. With $d^3\mathbf{y} = 4\pi r^2 \, dr$,

$$\begin{aligned} \mathcal{V}_g &= \frac{1}{2} \rho \int_0^b \left[-\frac{GM}{2b^3} (3b^2 - r^2) \right] 4\pi r^2 \, dr \\ &= -\frac{\pi \rho GM}{b^3} \left(b^5 - \frac{b^5}{5} \right) = -\frac{4\pi \rho GM b^2}{5} = -\frac{3}{5} \frac{GM^2}{b}, \end{aligned} \quad (4)$$

where the last step again uses $\rho = 3M/(4\pi b^3)$. Next we use the field-energy form $\mathcal{V}_g = -\frac{1}{8\pi G} \int_{\mathbb{R}^3} \|\nabla \phi\|^2 \, d^3\mathbf{y}$, for which the integral extends over all space. Here $\|\nabla \phi\|^2 = g_r^2$, and

$$\begin{aligned} \int_{\mathbb{R}^3} \|\nabla \phi\|^2 \, d^3\mathbf{y} &= 4\pi \left(\frac{G^2 M^2}{b^6} \int_0^b r^4 \, dr + G^2 M^2 \int_b^\infty \frac{dr}{r^2} \right) \\ &= 4\pi G^2 M^2 \left(\frac{1}{5b} + \frac{1}{b} \right) = \frac{24\pi}{5} \frac{G^2 M^2}{b}, \end{aligned} \quad (5)$$

so that

$$\mathcal{V}_g = -\frac{1}{8\pi G} \cdot \frac{24\pi}{5} \frac{G^2 M^2}{b} = -\frac{3}{5} \frac{GM^2}{b}, \quad (6)$$

in agreement with eq. (4). It is worth noticing how differently the two formulae distribute the same total: in eq. (5) five sixths of the field energy resides *outside* the sphere, whereas eq. (4) has no exterior contribution at all. The two expressions differ by a divergence, and only their integrals agree.

For the Earth, with the values quoted in the preamble, $GM^2/b = 3.73 \times 10^{32} \text{ J}$ and hence

$$\mathcal{V}_g = -\frac{3}{5} \frac{GM^2}{b} \approx -2.24 \times 10^{32} \text{ J}. \quad (7)$$

For comparison, the same data give a mean density $3M/(4\pi b^3) \approx 5510 \text{ kg m}^{-3}$ and a surface gravity $GM/b^2 \approx 9.82 \text{ m s}^{-2}$. The energy radiated by the largest recorded earthquakes, such as the 1960 Chile event, is of order 10^{18} – 10^{19} J , so the gravitational binding energy exceeds the energy of the largest earthquake by a factor of 10^{13} – 10^{14} . Because the real Earth's mass is concentrated towards its centre, the true binding energy is somewhat larger in magnitude than the homogeneous estimate. The comparison makes plain why gravity cannot be ignored in the long-period dynamics: even a minute fractional change in the gravitational energy is enormous on the scale of seismic energies.

Example 2: Referential gravity under a rigid motion. Let φ be a motion of the body, and let $\tilde{\varphi} = \mathbf{a} + \mathbf{Q}\varphi$ be the motion obtained by composing it with a rigid translation $\mathbf{a}$ and rotation $\mathbf{Q}$ (so that $Q_{ki}Q_{kj} = \delta_{ij}$). Show directly from the integral expressions of the lecture that the referential gravitational potential is unchanged, $\tilde{\zeta} = \zeta$, while the referential gravitational acceleration transforms as $\tilde{\gamma}_i = Q_{ij}\gamma_j$.

Comment on the significance of these results for the Lagrangian and for the equations of motion in a rotating frame.

Solution. The referential potential is given in the lecture by

$$\zeta(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')}{\|\varphi(\mathbf{x}, t) - \varphi(\mathbf{x}', t)\|} d^3\mathbf{x}', \quad (8)$$

and depends on the motion only through the separation $\varphi(\mathbf{x}, t) - \varphi(\mathbf{x}', t)$ of pairs of material points. Under the rigid motion this separation becomes

$$\tilde{\varphi}(\mathbf{x}, t) - \tilde{\varphi}(\mathbf{x}', t) = \mathbf{a} + \mathbf{Q}\varphi(\mathbf{x}, t) - \mathbf{a} - \mathbf{Q}\varphi(\mathbf{x}', t) = \mathbf{Q} [\varphi(\mathbf{x}, t) - \varphi(\mathbf{x}', t)], \quad (9)$$

so the translation cancels identically. Writing $\mathbf{d} = \varphi(\mathbf{x}, t) - \varphi(\mathbf{x}', t)$, orthogonality of $\mathbf{Q}$ gives

$$\|\mathbf{Q}\mathbf{d}\|^2 = Q_{ki}d_i Q_{kj}d_j = \delta_{ij}d_i d_j = \|\mathbf{d}\|^2, \quad (10)$$

and hence every denominator in eq. (8) is unchanged. The referential density $\rho(\mathbf{x}')$ is a property of the reference body and does not depend on the motion at all. It follows at once that $\tilde{\zeta}(\mathbf{x}, t) = \zeta(\mathbf{x}, t)$.

The referential acceleration is

$$\gamma_i(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')d_i}{\|\mathbf{d}\|^3} d^3\mathbf{x}', \quad (11)$$

with $\mathbf{d}$ as above. Under the rigid motion $d_i \rightarrow Q_{ij}d_j$ in the numerator while, by eq. (10), the denominator is unchanged. Since $\mathbf{Q}$ does not depend on $\mathbf{x}'$ it can be taken outside the integral, and so

$$\tilde{\gamma}_i(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')Q_{ij}d_j}{\|\mathbf{d}\|^3} d^3\mathbf{x}' = Q_{ij}\gamma_j(\mathbf{x}, t), \quad (12)$$

as required.

The same conclusions can be reached from the spatial description. If the spatial density of the original motion is $\varrho(\mathbf{y}, t)$, that of the rigidly moved body is $\tilde{\varrho}(\mathbf{y}, t) = \varrho[\mathbf{Q}^T(\mathbf{y} - \mathbf{a}), t]$, and because the Laplacian is invariant under rigid motions the solution of Poisson's equation is simply carried along, $\tilde{\phi}(\mathbf{y}, t) = \phi[\mathbf{Q}^T(\mathbf{y} - \mathbf{a}), t]$. Evaluating at $\mathbf{y} = \tilde{\varphi}(\mathbf{x}, t)$ gives $\tilde{\zeta}(\mathbf{x}, t) = \phi[\varphi(\mathbf{x}, t), t] = \zeta(\mathbf{x}, t)$, while the chain rule applied to $\tilde{g}_i = -\partial\tilde{\phi}/\partial y_i$ produces the factor Q_{ij} .

Two consequences are worth drawing out. First, the gravitational binding energy $\mathcal{V}_g = \frac{1}{2} \int_M \rho \zeta d^3\mathbf{x}$ is unchanged by a rigid motion. The Lagrangian of the lecture is therefore frame indifferent in the same sense as the elastic strain energy, which is essential if the equations of motion are to be physically meaningful. Second, taking $\mathbf{a} = \mathbf{\Phi}(t)$ and $\mathbf{Q} = \mathbf{R}(t)$ to be the translation and rotation introduced in the lecture for the co-rotating frame, eq. (12) is precisely the identity $\gamma_i = R_{ij}\bar{\gamma}_j$ used there, with $\bar{\gamma}$ computed from the relative motion $\bar{\varphi}$. Gravity, unlike the inertial terms, is oblivious to the rotation of the frame; it is only the acceleration that acquires Coriolis and centrifugal contributions.

Example 3: A steadily rotating frame. Let $\mathbf{R}(t)$ be the rotation through angle Ωt about the $\hat{\mathbf{e}}_3$ axis. Compute the matrix $R_{kj} \frac{dR_{kl}}{dt}$, verify that it is anti-symmetric, and use the lecture's relation $R_{kj} \frac{dR_{kl}}{dt} = \epsilon_{jml} \Omega_m$ to identify the angular velocity vector $\boldsymbol{\Omega}$. Write out the Coriolis and centrifugal accelerations in components. Show that the centrifugal acceleration $\epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m$ is the gradient of $\psi = -\frac{1}{2} \Omega^2 (x_1^2 + x_2^2)$, and check this against the general expression $\psi = \frac{1}{2} (\Omega_i \Omega_j - \Omega_k \Omega_k \delta_{ij}) x_i x_j$ valid for an arbitrary rotation vector. Evaluate the centrifugal acceleration at the Earth's equator.

Solution. Writing $c = \cos \Omega t$ and $s = \sin \Omega t$, the rotation matrix and its time derivative are

$$\mathbf{R}(t) = \begin{pmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \frac{d\mathbf{R}}{dt} = \Omega \begin{pmatrix} -s & -c & 0 \\ c & -s & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (13)$$

The quantity $R_{kj} \frac{dR_{kl}}{dt}$ is the (j, l) element of $\mathbf{R}^T \frac{d\mathbf{R}}{dt}$, and multiplying out,

$$\begin{aligned} \mathbf{R}^T \frac{d\mathbf{R}}{dt} &= \Omega \begin{pmatrix} c & s & 0 \\ -s & c & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -s & -c & 0 \\ c & -s & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= \Omega \begin{pmatrix} -cs + sc & -c^2 - s^2 & 0 \\ s^2 + c^2 & sc - cs & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\Omega & 0 \\ \Omega & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \end{aligned} \quad (14)$$

which is manifestly anti-symmetric and, notably, independent of time. To identify $\boldsymbol{\Omega}$ we compare with $\epsilon_{jml} \Omega_m$. Taking $(j, l) = (1, 2)$ gives $\epsilon_{1m2} \Omega_m = \epsilon_{132} \Omega_3 = -\Omega_3$, which must equal $-\Omega$, while $(j, l) = (2, 3)$ gives $\epsilon_{2m3} \Omega_m = \Omega_1 = 0$ and $(j, l) = (3, 1)$ gives $\Omega_2 = 0$. Hence $\boldsymbol{\Omega} = \Omega \hat{\mathbf{e}}_3$, as expected for a rotation about the third axis. More generally, contracting the lecture's relation with ϵ_{jnl} and using $\epsilon_{jml} \epsilon_{jnl} = 2\delta_{mn}$ gives the explicit formula

$$\Omega_n = \frac{1}{2} \epsilon_{jnl} R_{kj} \frac{dR_{kl}}{dt}, \quad (15)$$

which recovers the angular velocity from any rotation history $\mathbf{R}(t)$.

With $\boldsymbol{\Omega} = \Omega \hat{\mathbf{e}}_3$ the Coriolis acceleration $2\epsilon_{ijk} \Omega_j \bar{v}_k$ is $2\boldsymbol{\Omega} \times \bar{\mathbf{v}}$, with components

$$2\epsilon_{ijk} \Omega_j \bar{v}_k = 2\Omega (-\bar{v}_2, \bar{v}_1, 0), \quad (16)$$

so it is perpendicular to both the rotation axis and the relative velocity, and vanishes for motion parallel to the axis. For the centrifugal term we use the identity $\epsilon_{ijk} \epsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}$ to write

$$\epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m = \Omega_i \Omega_j x_j - \Omega_j \Omega_j x_i = [\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x})]_i, \quad (17)$$

and for $\boldsymbol{\Omega} = \Omega \hat{\mathbf{e}}_3$ this is

$$\epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m = \Omega^2 (0, 0, x_3) - \Omega^2 (x_1, x_2, x_3) = -\Omega^2 (x_1, x_2, 0), \quad (18)$$

which points towards the rotation axis with magnitude Ω^2 times the distance from it.

Now let $\psi = -\frac{1}{2}\Omega^2(x_1^2 + x_2^2)$. Its gradient is $\nabla\psi = -\Omega^2(x_1, x_2, 0)$, which by eq. (18) is exactly the centrifugal acceleration. Thus the term appearing on the left-hand side of the lecture's equations of motion is $\rho \partial\psi/\partial x_i$, and when it is carried to the right-hand side it becomes the centrifugal body force $-\rho \partial\psi/\partial x_i$ per unit volume, directed away from the axis; ψ is the **centrifugal potential**. To check the general formula, differentiate $\psi = \frac{1}{2}(\Omega_i\Omega_j - \Omega_k\Omega_k\delta_{ij})x_ix_j$, noting that the coefficient matrix is symmetric, to find

$$\frac{\partial\psi}{\partial x_i} = (\Omega_i\Omega_j - \Omega_k\Omega_k\delta_{ij})x_j = \Omega_i\Omega_jx_j - \Omega_k\Omega_kx_i, \quad (19)$$

which is eq. (17). Substituting $\mathbf{\Omega} = \Omega\hat{\mathbf{e}}_3$ into the general formula gives

$$\psi = \frac{1}{2} \left[\Omega^2x_3^2 - \Omega^2(x_1^2 + x_2^2 + x_3^2) \right] = -\frac{1}{2}\Omega^2(x_1^2 + x_2^2), \quad (20)$$

in agreement with the special case. Note that $\nabla^2\psi = -2\Omega^2 \neq 0$, so unlike the gravitational potential in free space the centrifugal potential is not harmonic.

At the equator the distance from the axis is b , and the centrifugal acceleration has magnitude

$$\Omega^2b = (7.29 \times 10^{-5})^2 \times 6.371 \times 10^6 \text{ m s}^{-2} \approx 3.39 \times 10^{-2} \text{ m s}^{-2}, \quad (21)$$

which is $\Omega^2b^3/GM \approx 3.4 \times 10^{-3}$, or about one part in 290, of the surface gravity found in Example 1. This small ratio is what permits the Earth's equilibrium figure to be treated as a slightly flattened sphere in the following lecture, where the centrifugal and gravitational potentials are combined into a single geopotential.

Example 4: Scaling analysis with numbers. Using the order-of-magnitude estimates of the lecture, with $\bar{\rho} = 5000 \text{ kg m}^{-3}$ and $\bar{v} = 8000 \text{ m s}^{-1}$ as typical density and wave speed, evaluate the ratio of gravitational to elastic forces, the ratio ΩT of Coriolis to inertial terms, and the ratio of centrifugal to inertial terms, for (a) a 1 Hz body wave with $L \sim 10 \text{ km}$; (b) a 100 s surface wave with $L \sim 400 \text{ km}$; and (c) the gravest spheroidal mode ${}_0S_2$, with period 54 min and $L \sim b$. Tabulate the results and state where gravity and rotation matter.

Solution. From the lecture, the ratios of the various terms in the linearised equations of motion to the elastic and inertial terms are

$$\frac{\text{gravitational}}{\text{elastic}} = \frac{4\pi G\bar{\rho}L^2}{\bar{v}^2}, \quad \frac{\text{Coriolis}}{\text{inertial}} = \Omega T, \quad \frac{\text{centrifugal}}{\text{inertial}} = \frac{2\bar{\rho}\Omega^2U}{\bar{\rho}U/T^2} = 2(\Omega T)^2, \quad (22)$$

where the final ratio follows from the lecture's estimate $2\bar{\rho}\Omega^2U$ for the centrifugal term; the factor of two is an upper bound, attained when $\mathbf{u}$ is perpendicular to $\mathbf{\Omega}$, and is of no consequence in an order-of-magnitude argument. It is convenient to note the combination

$$4\pi G\bar{\rho} = 4\pi \times 6.674 \times 10^{-11} \times 5000 \text{ s}^{-2} = 4.19 \times 10^{-6} \text{ s}^{-2}, \quad (23)$$

so that the gravitational ratio is $(L/L_g)^2$ with $L_g = \bar{v}/\sqrt{4\pi G\bar{\rho}} \approx 3900 \text{ km}$, while the rotational ratios involve T compared with $\Omega^{-1} \approx 1.37 \times 10^4 \text{ s} \approx 3.8 \text{ h}$. Substituting the three sets of values gives the following table.

Case	T (s)	L (km)	grav./elastic	ΩT	$2(\Omega T)^2$
(a) body wave	1	10	6.55×10^{-6}	7.29×10^{-5}	1.06×10^{-8}
(b) surface wave	100	400	1.05×10^{-2}	7.29×10^{-3}	1.06×10^{-4}
(c) ${}_0S_2$	3240	6371	2.66	0.236	0.112

For the body wave every correction is utterly negligible: gravity enters at the level of a few parts per million and rotation at the level of one part in 10^4 or below, so the elastodynamics of the earlier lectures is exact for all practical purposes. For the 100 s surface wave the gravitational correction has grown to about one per cent, as expected from the L^2 scaling, and although still small it is comparable to the fractional variations in wave speed that surface-wave tomography seeks to resolve, so gravity is routinely included in surface-wave calculations at these periods. Rotation, by contrast, remains below one per cent. For ${}_0S_2$ the gravitational ratio is of order unity – this is the lecture’s estimate of about 3 – and gravity is as important as elasticity; indeed, the restoring force for this mode is largely gravitational. The Coriolis ratio $\Omega T \approx 0.24$ is no longer small, and it is responsible for the observable splitting of the gravest modes into multiplets. The centrifugal ratio $2(\Omega T)^2 \approx 0.1$ is smaller but not negligible at this period.

The table illustrates the general conclusion of the lecture. Gravity matters for deformations whose length scale approaches L_g of a few thousand kilometres, which for seismic waves means periods of several hundred seconds and longer, while rotation matters for time scales that approach Ω^{-1} , i.e. for the longest-period normal modes, and is dominant only for phenomena such as tides with periods comparable to a day.

Example 5: Linearised gravitational acceleration for rigid displacements. The lecture gives the first-order perturbation to the referential gravitational acceleration as

$$\gamma_i^1 = -G \int_M \rho' \left(\frac{\delta_{ij}}{\|\mathbf{x} - \mathbf{x}'\|^3} - \frac{3(x_i - x'_i)(x_j - x'_j)}{\|\mathbf{x} - \mathbf{x}'\|^5} \right) (u_j - u'_j) d^3 \mathbf{x}'. \quad (24)$$

Show that a rigid translation $\mathbf{u} = \mathbf{u}_0$ gives $\gamma^1 = \mathbf{0}$, and that an infinitesimal rigid rotation $\mathbf{u} = \boldsymbol{\theta} \times \mathbf{x}$, with $\boldsymbol{\theta}$ a constant vector, gives $\gamma^1 = \boldsymbol{\theta} \times \gamma^0$, where γ^0 is the equilibrium referential acceleration. Explain why both results are to be expected.

Solution. For the translation, $u_j - u'_j = u_{0j} - u_{0j} = 0$ for every pair of points, so the integrand in eq. (24) vanishes identically and $\gamma^1 = \mathbf{0}$. Nothing needs to be assumed about the density or the shape of the body. Physically, $\gamma_i(\mathbf{x}, t) = g_i[\boldsymbol{\varphi}(\mathbf{x}, t), t]$ is the acceleration felt by the material particle labelled $\mathbf{x}$. Translating the whole body translates its gravitational field along with it, and each particle continues to feel exactly the field it felt before; only in the spatial description, at fixed $\mathbf{y}$, does the field change. This is the linearised form of the translational invariance established in Example 2.

For the rotation write $\mathbf{r} = \mathbf{x} - \mathbf{x}'$ and $r = \|\mathbf{r}\|$. Then

$$u_j - u'_j = \epsilon_{jkl} \theta_k (x_l - x'_l) = \epsilon_{jkl} \theta_k r_l = (\boldsymbol{\theta} \times \mathbf{r})_j, \quad (25)$$

which is perpendicular to $\mathbf{r}$. Contracting with the kernel in eq. (24),

$$\left(\frac{\delta_{ij}}{r^3} - \frac{3r_i r_j}{r^5} \right) (\boldsymbol{\theta} \times \mathbf{r})_j = \frac{(\boldsymbol{\theta} \times \mathbf{r})_i}{r^3} - \frac{3r_i \mathbf{r} \cdot (\boldsymbol{\theta} \times \mathbf{r})}{r^5} = \frac{(\boldsymbol{\theta} \times \mathbf{r})_i}{r^3}, \quad (26)$$

because $\mathbf{r} \cdot (\boldsymbol{\theta} \times \mathbf{r}) = 0$. The second, traceless part of the kernel therefore contributes nothing, and

$$\gamma_i^1 = -G \int_M \rho' \frac{\epsilon_{ikl} \theta_k r_l}{r^3} d^3 \mathbf{x}' = \epsilon_{ikl} \theta_k \left(-G \int_M \rho' \frac{x_l - x'_l}{\|\mathbf{x} - \mathbf{x}'\|^3} d^3 \mathbf{x}' \right) = \epsilon_{ikl} \theta_k \gamma_l^0, \quad (27)$$

where the bracketed integral is the equilibrium referential acceleration obtained by setting $\varphi(\mathbf{x}) = \mathbf{x}$ in the lecture's integral formula. That is, $\gamma^1 = \boldsymbol{\theta} \times \gamma^0$.

This too is what Example 2 predicts. A finite rotation $\mathbf{Q}$ transforms the referential acceleration as $\gamma \rightarrow \mathbf{Q}\gamma$, and for an infinitesimal rotation $Q_{ij} = \delta_{ij} + s \epsilon_{ikj} \theta_k + O(s^2)$, corresponding to the motion $\varphi_i = x_i + s \epsilon_{ikj} \theta_k x_j + O(s^2)$, so that $\gamma_i = \gamma_i^0 + s \epsilon_{ikj} \theta_k \gamma_j^0 + O(s^2)$; the first-order term is exactly eq. (27). The gravitational field is rigidly rotated with the body, and each particle feels the rotated field. In the non-rotating case, Example 2 shows that a rigid motion of an equilibrium configuration is again an equilibrium configuration, so rigid translations and rotations must be zero-frequency solutions of the linearised equations of motion; the two calculations here are the gravitational part of that statement, and the remaining terms must conspire in the same way. When the frame rotates the argument no longer applies as it stands, because the centrifugal term $\epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m$ does not vanish for a rigid translation transverse to the axis, and the rigid-body modes of a rotating planet require separate treatment.